Software Design Document

<Sustainable Shopping Assistant>

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Summery

The Sustainable Shopping Assistant is a web-based platform developed to help consumers analyze, track, and optimize the environmental impact of their online purchases. With the rapid expansion of e-commerce and increasing corporate transparency issues (such as greenwashing), there is a critical need for automated systems that can evaluate the sustainability claims of retail products.

This application utilizes advanced Natural Language Processing (NLP) techniques, web scraping modules, and a fine-tuned Transformer-based deep learning model (RoBERTa) to analyze product descriptions, material composition, and supply chain logistics data. By parsing these textual and structural datasets, the system calculates multi-criteria environmental impact metrics and provides real-time eco-scores along with a confidence rating.

The application architecture relies on a highly responsive frontend built using ReactJS, HTML5, CSS3, and JavaScript, paired with a high-performance backend powered by FastAPI (Python). Data tracking, user carbon offset histories, and reward tokens are managed using a secure MongoDB database schema. The proposed system empowers users, eco-conscious organizations, and research communities to foster sustainable consumption and hold retail brands accountable.

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1. Introduction

The **Sustainable Shopping Assistant** is an AI-powered software application engineered to assist modern consumers in evaluating, tracking, and optimizing the ecological impact of their e-commerce purchases. With the unprecedented growth of retail logistics and fast fashion, consumers are increasingly demanding transparency regarding the carbon footprint, material composition, and ethical sourcing of retail products.

However, corporate marketing often hides severe environmental impacts under vague assertions—a practice widely known as **greenwashing**. Consumers typically lack the data consolidation tools required to make immediate eco-friendly decisions at the point of sale.

This system bridges the gap by leveraging automated web scraping, Natural Language Processing (NLP), and fine-tuned deep learning models to parse unstructured product data from online stores. By converting complex text descriptions and supply chain parameters into unified environmental metrics, the application provides real-time sustainability grades and recommends eco-friendly alternatives.

Objectives

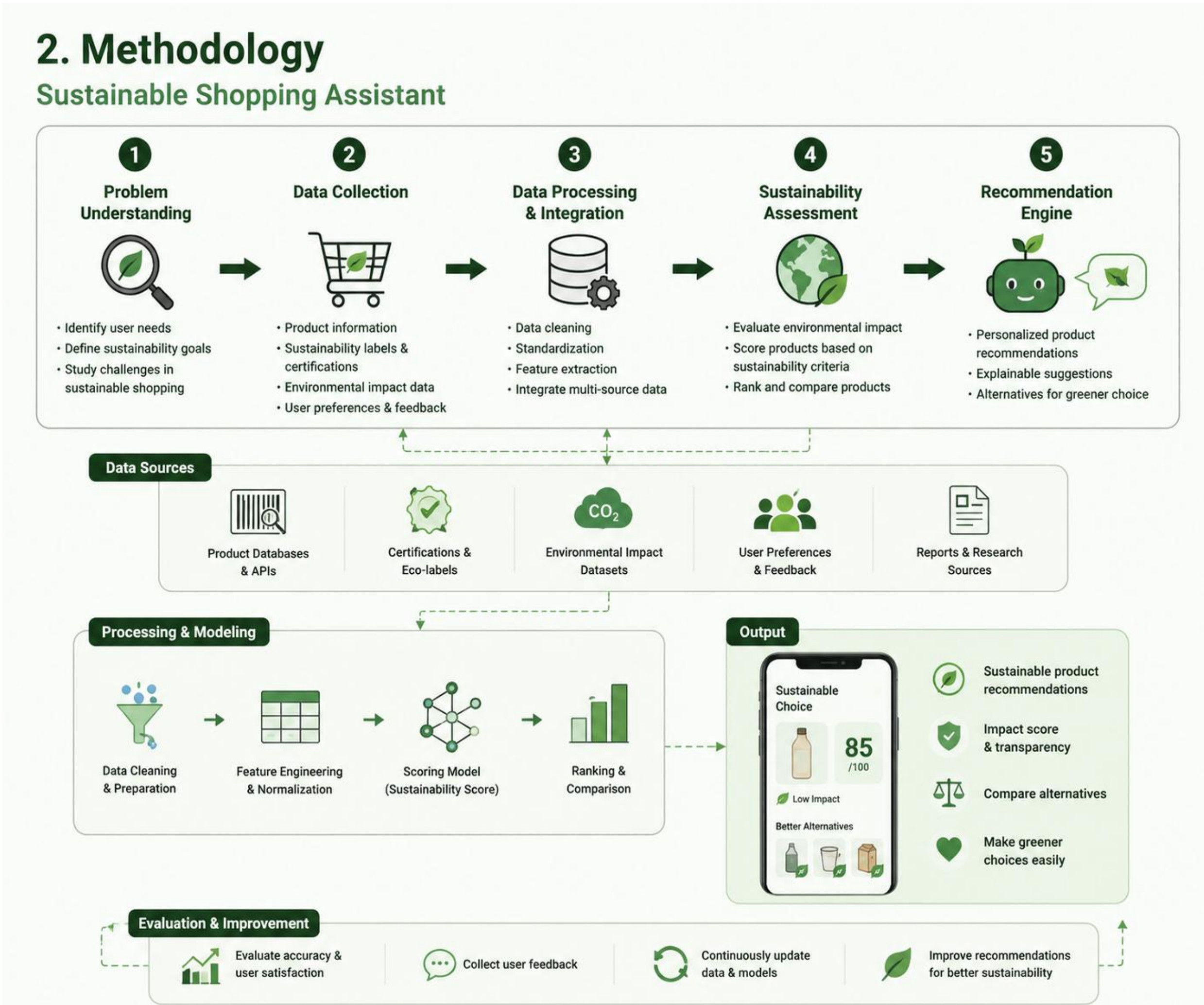
1. **Promote Eco-Conscious Shopping:** Empower users with transparent environmental impact scores for everyday products.
2. **Carbon Footprint Tracking:** Calculate and display estimated carbon emissions associated with product manufacturing and logistics.
3. **Alternative Recommendations:** Provide real-time suggestions for alternative, highly sustainable products.
4. **Green History Management:** Store user shopping behaviors and award "Eco-Credits" to incentivize sustainable choices.

Intuitive Dashboard: Provide users and administrators with comprehensive data visualization regarding green impact trends.

Technologies Used

Component	Technology
Frontend	ReactJS, HTML5, CSS3, JavaScript (ES6)
Backend Framework	FastAPI (Python)
AI/ML Scoring Model	RoBERTa / Custom Multi-Criteria Decision Model (MCDM)
Database	MongoDB (NoSQL)
Scraping & Data Processing	Beautiful Soup, spaCy, NLTK
Development Environment	Visual Studio Code

2. Methodology



2.1 Dataset Collection

The system operates on an aggregated data retrieval strategy, capturing manufacturing descriptors, shipping coordinates, and material sustainability properties from open-source indices and live web endpoints.

Table 2.1: Dataset Collection & Sources Breakdown

Source / Repository	Data Classification Category	Intent & Processing Purpose
Kaggle Retail Datasets	E-Commerce Data Matrices	Supplies standardized baseline product text fields across global store brands.
Open Environmental Logs	Sustainability Benchmarks	Provides reliable metrics on materials, logistics routing, and carbon emissions.
Custom Scraped Link Targets	Brand Product Descriptions	Live web scraping data containing real marketing content for text categorization.
Eco-Sourcing Blogs & Reports	Ethical Consumer Reviews	Provides conversational sentiment records and independent brand analysis.

2.2 Data Preprocessing

Raw scraped e-commerce textual attributes are highly unstructured and contain nested HTML elements, noisy advertising terms, and redundant code syntax. In this layer, script frameworks normalize all textual targets by isolating structural strings from web metadata tags. Text fields undergo lowercasing, sequence adjustments, and preliminary filtering to prepare the attributes for contextual analysis.

2.3 Data Cleaning

Data cleaning isolates critical data markers from conversational linguistic filler. Using the **Natural Language Toolkit (NLTK)**, the system performs tokenization (breaking text blocks into distinct string arrays), filters out standard non-analytical stop-words (e.g., "the", "and", "is"), and applies lemmatization to reduce variance across corporate descriptions.

2.4 Stylometric Feature Analysis

To uncover corporate greenwashing, the system maps the stylistic structure of product advertisements. It extracts text density metrics, structural patterns, and specific buzzword frequencies (such as "natural," "pure," or "eco-safe") that lack verifiable certification links. By evaluating token ratios and language transparency indices, the framework creates a baseline to differentiate verified sustainability data from marketing exaggeration.

2.5 Model Application

The system feeds the processed text profiles and structural tokens into a specialized multi-layer evaluation pipeline. A fine-tuned **RoBERTa (Robustly Optimized BERT Pretraining Approach)** language model analyzes unstructured text to identify marketing biases, verify corporate transparency, and classify the overall reliability of brand claims. Simultaneously, a quantitative **Multi-Criteria Decision Model (MCDM)** processes material parameters to calculate the final product score:

$$\text{Eco-Score} = (0.50 \times \text{Material Score}) + (0.30 \times \text{Carbon Footprint Score}) + (0.20 \times \text{Packaging Score})$$

2.6 Dataset Splitting

To optimize text classification accuracy and minimize error matrices, the foundational linguistic reference datasets are split into three independent testing buckets:

Table 2.2: Dataset Split Matrix

Dataset Type	Percentage	Purpose
Training Data	70%	Extracted tokens trained to learn deep textual linguistic parameters.
Validation Data	15%	Hyperparameter optimization checking precision loss curves.
Testing Data	15%	Final metric evaluation recording model target accuracy.

2.7 Training Phase

The deep learning pipeline runs on dedicated host clusters utilizing optimized gradient updates to map material contexts to sustainability scores.

Table 2.3: Training Phase Parameters Configuration

Hyperparameter Category	Configured Assessment Value	Analytical Purpose
Training Epochs Range	5 – 10	Manages gradient iterations to ensure full text pattern mastery without overfitting.
Processing Batch Size	16 / 32	Defines concurrent samples grouped to optimize RAM memory utilization.

Hyperparameter Category	Configured Assessment Value	Analytical Purpose
Stochastic Optimizer	AdamW	Updates network weights using decoupled weight decay formulas.
Initial Learning Rate	0.0001	Regulates training pace step sizing to locate optimal local minima.

2.8 Testing and Validation

Validation cycles run continually against hidden test arrays to record model accuracy, precision benchmarks, and recall metrics. This step prevents the system from misinterpreting authentic ecological assertions or failing to detect greenwashed advertising strings.

2.9 Workflow of Detection System

The complete sequential data flow—from the user inputting a link to the output of sustainability ratings and eco-friendly recommendations—is structured systematically within the platform pipeline.

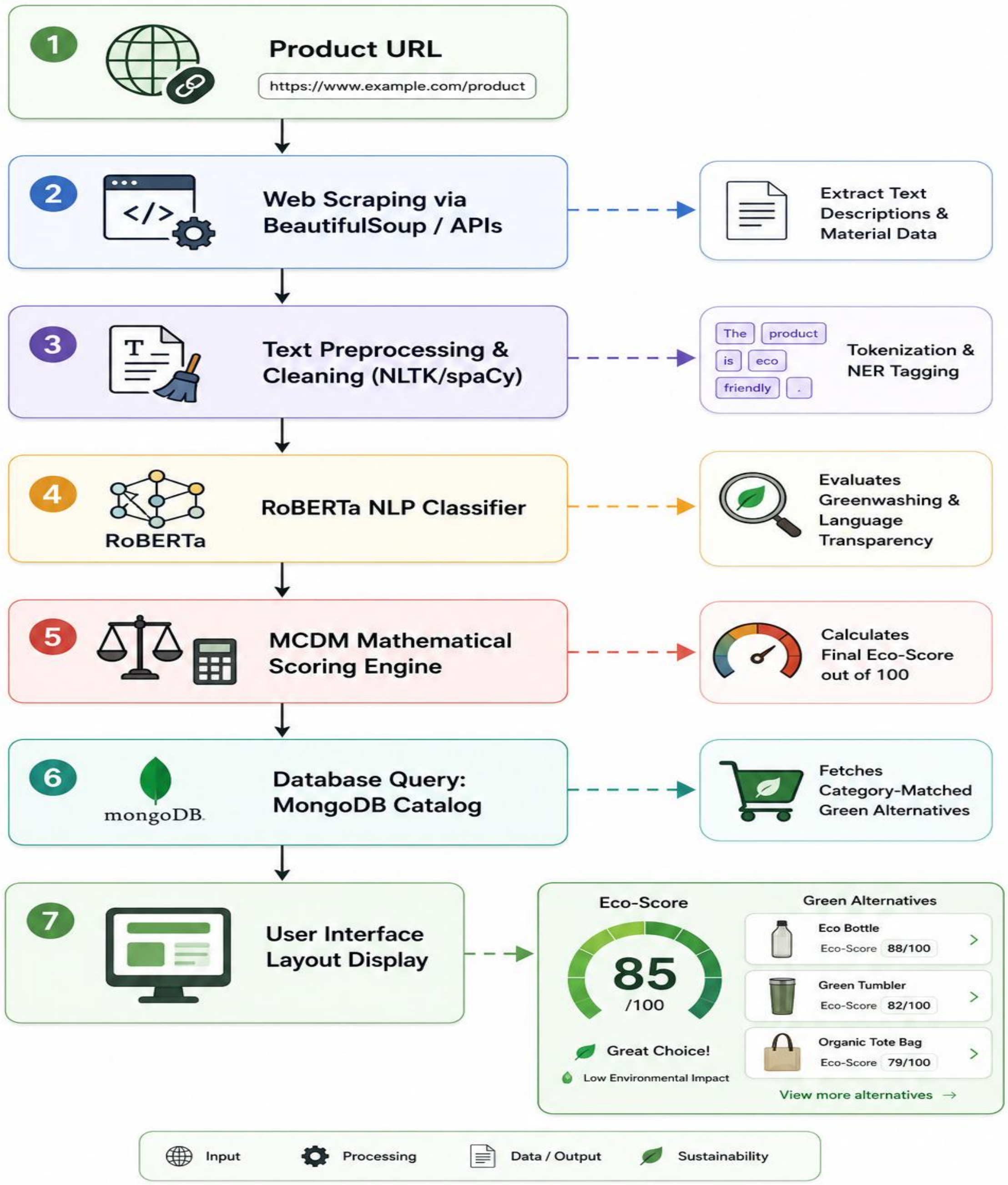


Figure 2.1: Workflow Data Pipeline Flowchart

3. Application Architecture

The application follows a three-tier architecture to decouple the presentation, application logic, and data storage layers, ensuring scalability and low-latency performance.

3.1 Frontend Architecture

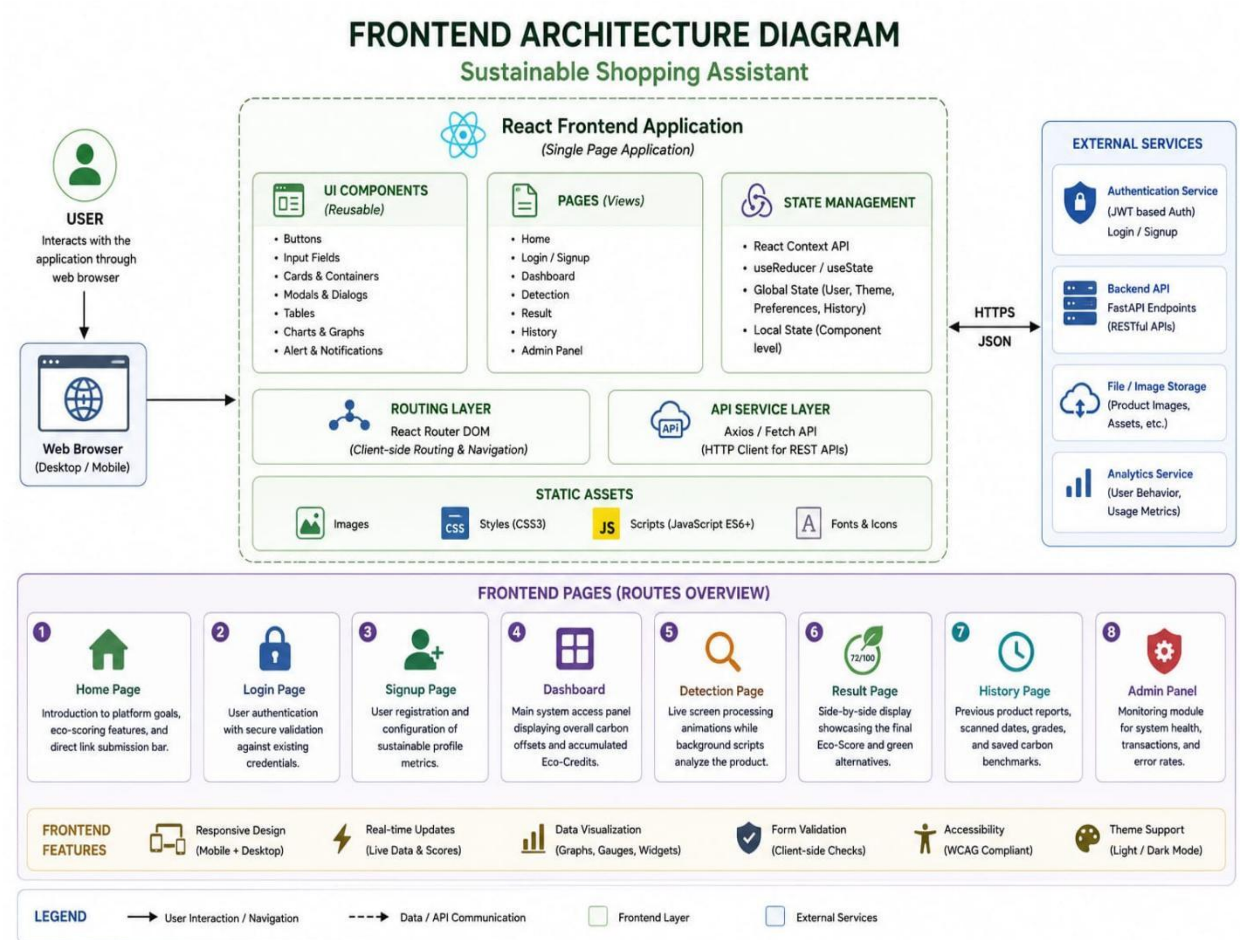
Technologies:

- HTML5
- CSS3
- JavaScript (ES6+)
- ReactJS (Component Framework)

Responsibilities:

1. **User Interaction:** Captures e-commerce product links, URLs, and clipboard text descriptors seamlessly.
2. **Product Input Handling:** Sanitizes user text fields and safely sends payloads to server gateways.
3. **Result Display:** Visualizes complex environmental scoring breakdowns using real-time graphics and color-coded alert widgets.
4. **Dashboard Interface:** Renders user carbon footprint metrics and environmental impact histories dynamically.
5. **History Management:** Provides sorting and filtering capabilities over previous product scanning records.

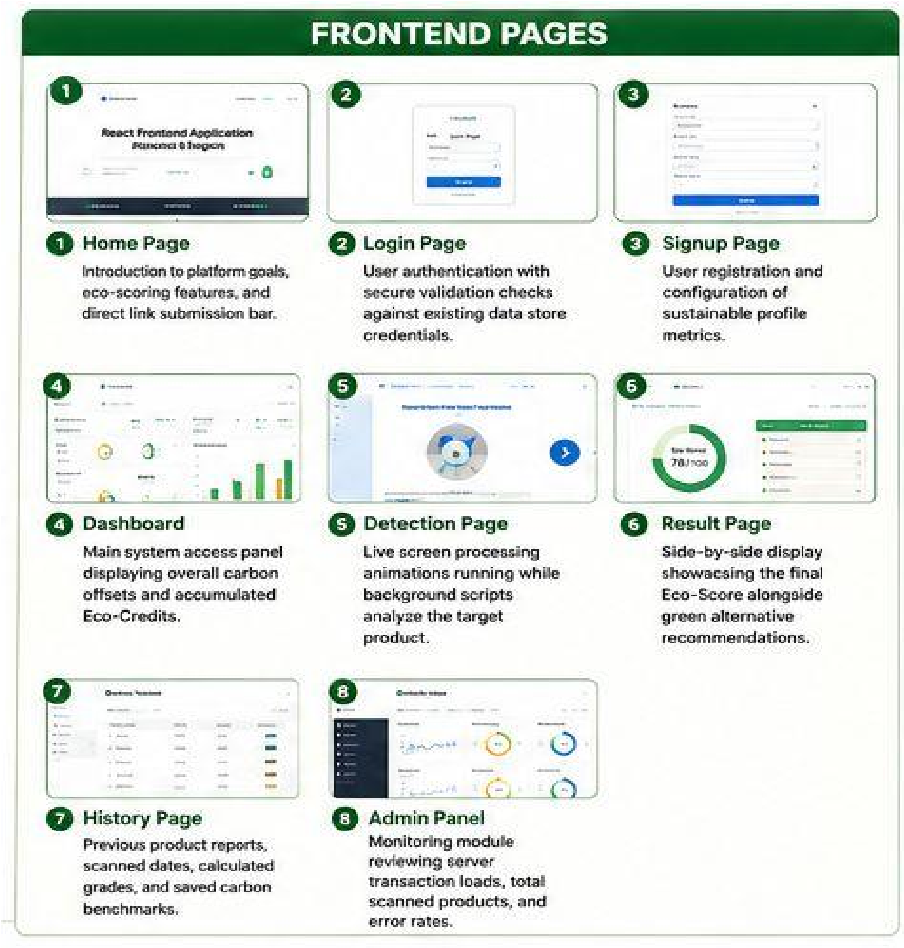
Frontend Architecture Diagram



Frontend Pages

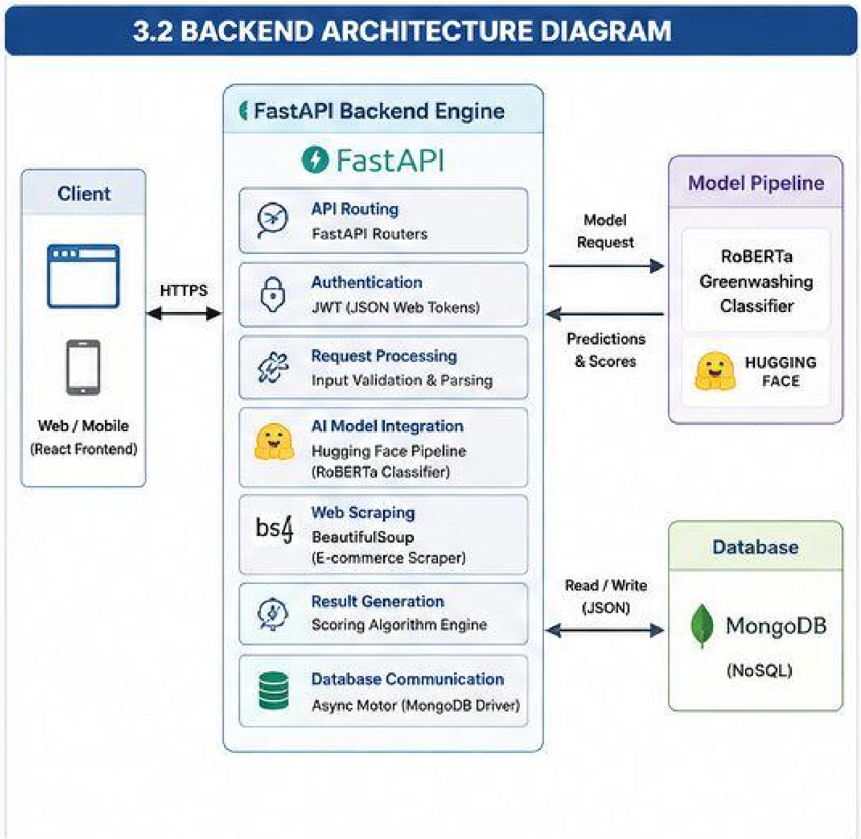
Page	Purpose
Home Page	Introduction to platform goals, eco-scoring features, and direct link submission bar.
Login Page	User authentication with secure validation checks against existing data store credentials.
Signup Page	User registration and configuration of sustainable profile metrics.

Page	Purpose
Dashboard	Main system access panel displaying overall carbon offsets and accumulated Eco-Credits.
Detection Page	Live screen processing animations running while background scripts analyze the target product.
Result Page	Side-by-side display showcasing the final Eco-Score alongside green alternative recommendations.
History Page	Previous product reports, scanned dates, calculated grades, and saved carbon benchmarks.
Admin Panel	Monitoring module reviewing server transaction loads, total scanned products, and error rates.



3.2 Backend Architecture

Backend Architecture Diagram



Technology

FastAPI (Python)

Backend Responsibilities:

1. **API Handling:** Managing asynchronous REST endpoints to ensure non-blocking task delivery queues.
2. **User Authentication:** Implementing stateless security access configurations using JSON Web Tokens (JWT).
3. **AI Model Integration:** Initializing Hugging Face pipeline wrappers to deploy the RoBERTa greenwashing classifier.
4. **Request Processing:** Parsing input URLs to execute targeted e-commerce scraping operations via BeautifulSoup.

5. **Database Communication:** Performing query transformations to push and pull JSON documents from the storage cluster.
6. **Result Generation:** Running weighted algorithms to consolidate carbon, material, and packaging metrics into a final score.

APIs

API Route	HTTP Method	Function
/api/v1/auth/signup	POST	Register a new eco-conscious user profile in the database.
/api/v1/auth/login	POST	Authenticate shopper credentials and return a secure JWT access token.
/api/v1/product/detect	POST	Analyze product text, run greenwashing evaluation pipelines, and return scores.
/api/v1/user/history	GET	Fetch previous eco-scoring history metrics matching the active user session.
/api/v1/admin/monitoring	GET	Admin monitoring of system-wide request logs, scraping counts, and performance health.

BACKEND APIS

API ROUTE	HTTP METHOD	FUNCTION
1 /api/v1/auth/signup	POST	Register a new eco-conscious user profile in the database.
2 /api/v1/auth/login	POST	Authenticate shopper credentials and return a secure JWT access token.
3 /api/v1/product/detect	POST	Analyze product text, run greenwashing evaluation pipelines, and return scores.
4 /api/v1/user/history	GET	Fetch previous eco-scoring history metrics matching the active user session.
5 /api/v1/admin/monitoring	GET	Admin monitoring of system-wide request logs, scraping counts, and performance health.

3.3 Database Architecture

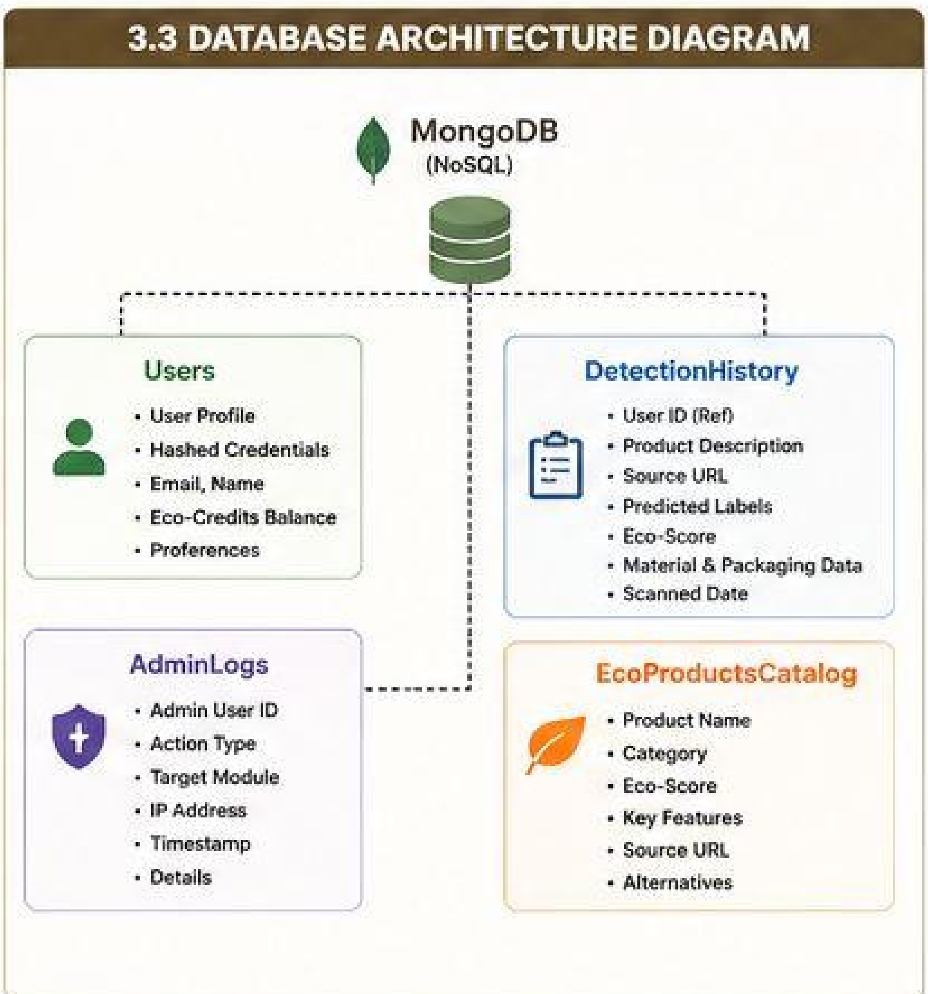
Technology

- MongoDB (NoSQL Document-Based Store)

Collections

Collection	Purpose
Users	Store secure user accounts, hashed security credentials, and accumulated Eco-Credits points balances.

Collection	Purpose
DetectionHistory	Store predictions, product descriptions, scraped source URLs, and individual Eco-Score results.
AdminLogs	System monitoring logs tracking configuration changes and secure administrator interactions.
EcoProductsCatalog	Cached index of verified sustainable replacement products utilized for alternative recommendations.



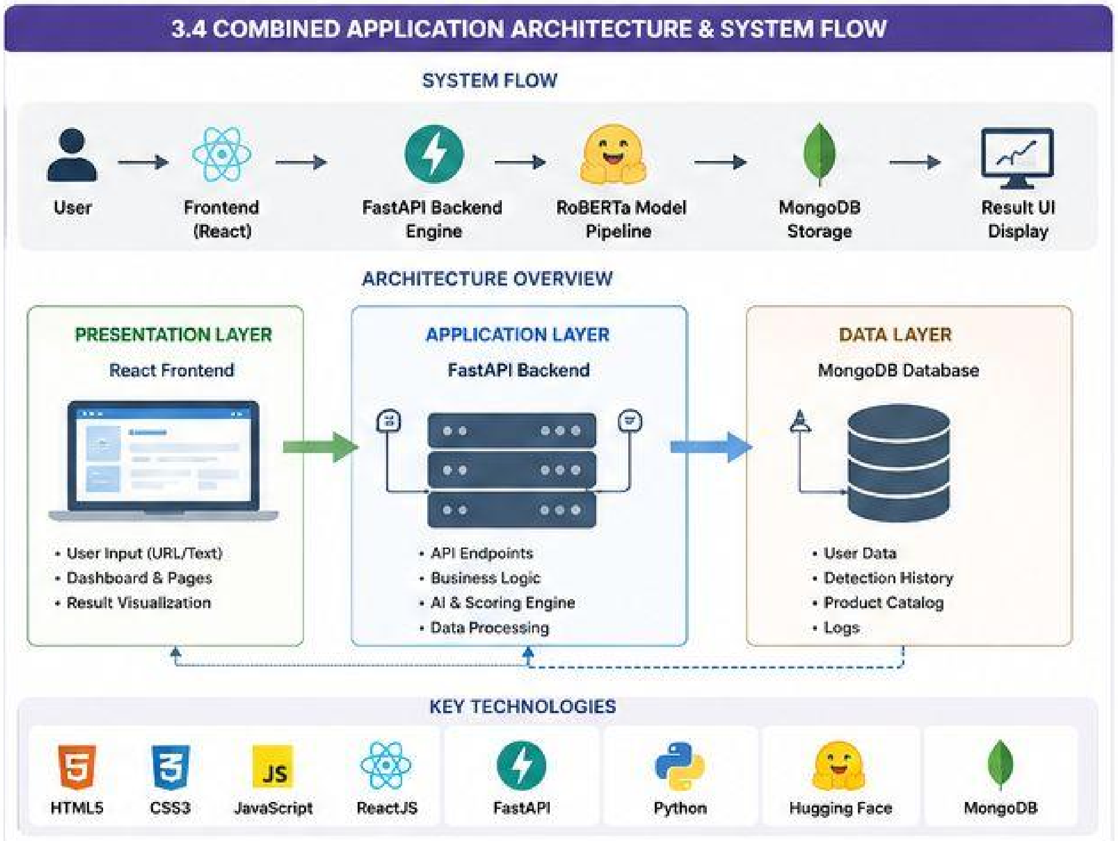
3.4 Combined Application Architecture

System Flow

$$\text{User} \longrightarrow \text{Frontend (HTML/CSS/JS/React)} \longrightarrow \text{FastAPI Backend Engine} \longrightarrow \text{RoBERTa Model Pipeline} \longrightarrow \text{MongoDB Storage} \longrightarrow \text{Result UI Display}$$

Architecture Explanation

1. **User submits text/URL through frontend:** The consumer inputs an e-commerce link or raw product text string into the React submission gateway.
2. **FastAPI receives request:** The high-performance backend routing layer captures the incoming payload and initiates a secured background worker process.
3. **Backend preprocesses text:** The system scrapes raw webpage elements, filters out noisy HTML scripts, and uses NLTK/spaCy pipelines to isolate core material tokens and brand claims.
4. **RoBERTa predicts text category:** The fine-tuned language model classifies text transparency to flag greenwashing, while multi-criteria decision algorithms calculate a quantitative score out of 100.
5. **Result stored in MongoDB:** The generated scoring matrix, material lists, and category tags are saved inside document collections indexed by user ID.
6. **Frontend displays output:** The client presentation layer updates dynamically to render a custom radial gauge of the product's Eco-Score alongside competitive green substitutes.

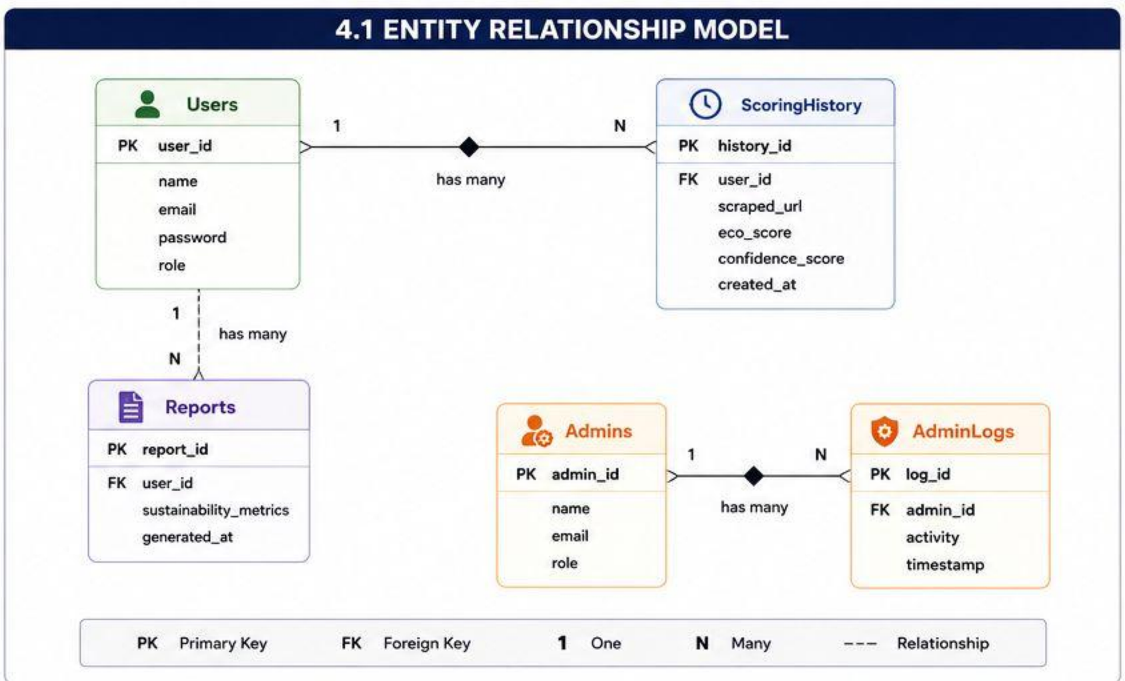

3.4 COMBINED APPLICATION ARCHITECTURE & SYSTEM FLOW
SYSTEM FLOW
User
Frontend (React)
FastAPI Backend Engine
RoBERTa Model Pipeline
MongoDB Storage
Result UI Display
ARCHITECTURE OVERVIEW
PRESENTATION LAYER
React Frontend
• User Input (URL/Text)
• Dashboard & Pages
• Result Visualization
APPLICATION LAYER
FastAPI Backend
• API Endpoints
• Business Logic
• AI & Scoring Engine
• Data Processing
DATA LAYER
MongoDB Database
• User Data
• Detection History
• Product Catalog
• Logs
KEY TECHNOLOGIES
HTML5
CSS3
JavaScript
ReactJS
FastAPI
Python
Hugging Face
MongoDB

4. Data Model Schema

The data layer utilizes a NoSQL document store pattern to accommodate varying product parameters, scraping targets, and real-time environmental metrics efficiently.

4.1 Entity Relationship Model

The system architecture coordinates logical interactions across consumer accounts, automated product assessments, green alternative mappings, and operational administrative track
entries.



Entities

User Entity

Attribute	Type	Description

Attribute	Type	Description
user_id	ObjectId	Unique identifier acting as the primary key.
name	String	Full name of the registered consumer.
email	String	Validated email address used for secure session logins.
password	String	Securely encrypted (Bcrypt-hashed) security string.
role	String	Authorization role string (e.g., "Shopper", "Admin").

ScoringHistory Entity **(Adapted from DetectionHistory)**

Attribute	Type	Description
history_id	ObjectId	Unique tracking identifier for each individual product scan.
user_id	ObjectId	Foreign key mapping back to the target user record.
scraped_url	Text	Absolute link target fetched for model calculations.
eco_score	String	Calculated environmental grade mapping (e.g., "78/100").
confidence_score	Float	Internal reliability index evaluating data text depth.
created_at	Date	Chronological system timestamp of the analysis run.

Reports Entity

Attribute	Type	Description
report_id	ObjectId	Unique primary key identifying compiled metrics files.
user_id	ObjectId	Foreign key associating the tracking sheets to a specific account.
sustainability_metrics	Text	Aggregated carbon footprint offsets and green shopping stats.
generated_at	Date	Precise generation timestamp for account statement updates.

AdminLogs Entity

Attribute	Type	Description
log_id	ObjectId	Sequential primary key entry sequence tracking admin actions.
admin_id	ObjectId	Foreign key referencing administrative management profiles.
activity	String	Text log tracking executed scraping configuration adjustments.
timestamp	Date	Live server clock log tracking structural data transactions.

Relationships

- **One User can have many Scoring Histories:** A single shopper profile can scan multiple e-commerce URLs over time, populating a personalized list of sustainability data entries.

- **One User can generate many Reports:** A registered consumer can output multiple recurring statements tracking their weekly carbon offsets and accumulated Eco-Credits.
- **One Admin can generate many Admin Logs:** System administrators can perform multiple structural backend tasks, with each action creating an entry in the system registry.

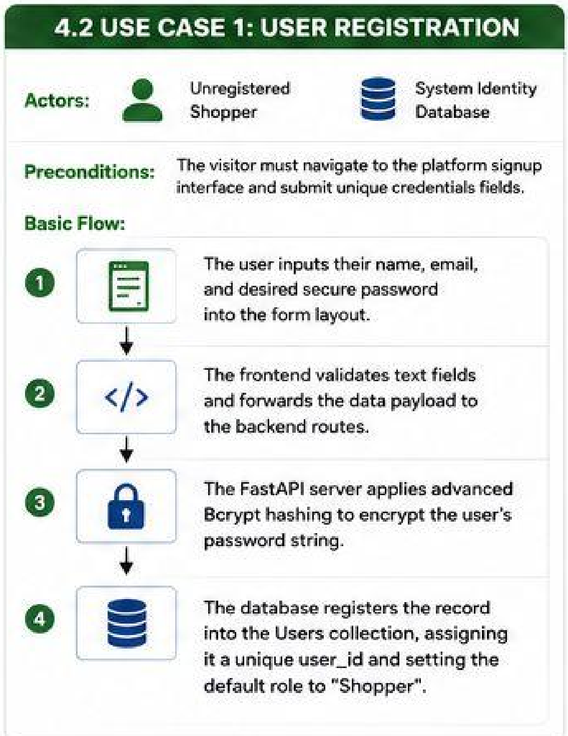
4.2 Use Case 1 (User Registration)

- **Actors:** Unregistered Shopper, System Identity Database.
- **Preconditions:** The visitor must navigate to the platform signup interface and submit unique credentials fields.

Basic Flow:

- The user inputs their name, email, and desired secure password into the form layout.
- The frontend validates text fields and forwards the data payload to the backend routes.
- The FastAPI server applies advanced Bcrypt hashing to encrypt the user's password string.

- The database registers the record into the **Users** collection, assigning it a unique user_id and setting the default role to "Shopper".



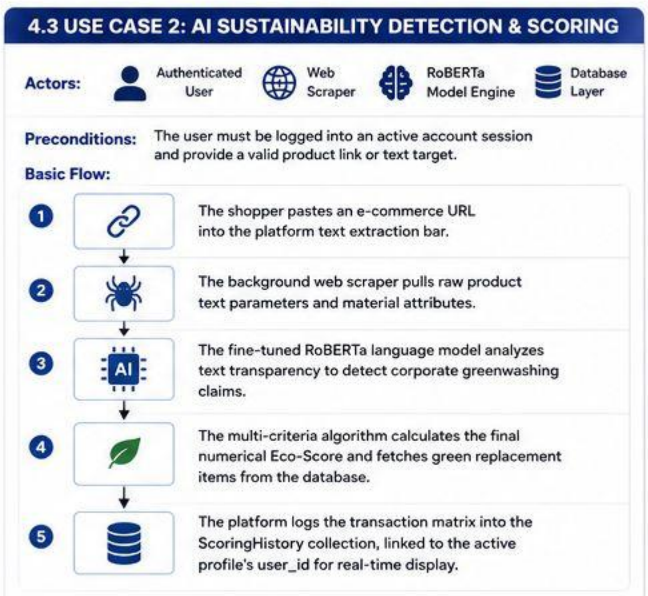
4.3 Use Case 2 (AI Sustainability Detection & Scoring)

- **Actors:** Authenticated User, Web Scraper, RoBERTa Model Engine, Database Layer.
- **Preconditions:** The user must be logged into an active account session and provide a valid product link or text target.

Basic Flow:

- The shopper pastes an e-commerce URL into the platform text extraction bar.
- The background web scraper pulls raw product text parameters and material attributes.

- The fine-tuned RoBERTa language model analyzes text transparency to detect corporate greenwashing claims.
- The multi-criteria algorithm calculates the final numerical Eco-Score and fetches green replacement items from the database.
- The platform logs the transaction matrix into the **ScoringHistory** collection, linked to the active profile's user_id for real-time display.



5. User Interface

5.1 Design Philosophy

The system follows a clean and modern academic design approach, combining a premium retail workspace aesthetic with clear data presentation panels.

Design Features:

1. **Minimal Layout:** Eliminates heavy backgrounds to keep the user's focus on product analysis scores and eco-alternative cards.
2. **Responsive Design:** Adapts fluidly across mobile browsers, tablets, and desktop setups for seamless shopping on any device.
3. **User-Friendly Navigation:** Simple navigation controls allow users to jump between live product scanning, dashboard tracking, and history data sheets instantly.
4. **Professional Academic Appearance:** Visualizes sustainability indicators through standardized tables and clean, research-driven data metrics.
5. **Lightweight Interface:** Optimizes code structure and visual assets to ensure fast page loads during active shopping tasks.

5.2 Color Scheme:

The interface applies a dedicated, sustainability-focused palette to distinguish eco-friendly components from carbon-intensive items visually.

Element	Color	Hex Code	Purpose

Element	Color	Hex Code	Purpose
Primary	Dark Forest Green	#1B5E20	Conveys environmental awareness across headers and top scores.
Secondary	Clean White	#FFFFFF	Acts as a minimalist canvas backdrop for data tables and forms.
Accent	Muted Mint Gray	#F1F8E9	Highlights product descriptive text blocks and alternative item lists.
Buttons	Vibrant Emerald	#4CAF50	Draws immediate attention to primary action and link submission fields.

5.3 Typography:

The design implements clear font configurations to ensure high readability for tabular material data and numerical analysis readouts.

Element	Font Family	Style / Application
Headings	Poppins	Clean, geometric sans-serif font applied to page headings and impact metrics.
Body Text	Arial	High-density font optimized for reading retail logs, material arrays, and lists.

5. USER INTERFACE

5.1 DESIGN PHILOSOPHY

Minimal Layout
Eliminates heavy backgrounds to keep the user's focus on product analysis scores and eco-alternative cards.

Responsive Design
Adapts fluidly across mobile browsers, tablets, and desktop setups for seamless shopping on any device.

User-Friendly Navigation
Simple navigation controls allow users to jump between live product scanning, dashboard tracking, and history data sheets instantly.

Professional Academic Appearance
Visualizes sustainability indicators through standardized tables and clean, research-driven data metrics.

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5.4 User Interface Screens

5.4.1 Home Page:

The landing view introduces the platform's core mission to stop corporate greenwashing and promote eco-conscious consumer habits.

Features:

- **Introduction:** Informative section explaining how the application uses AI to calculate carbon scores and evaluate packaging metrics.
- **Navigation Bar:** Global header giving instant routing access to the user Dashboard, Analysis History log, and account profile settings.
- **System Overview:** Step-by-step visual summary showing how simple URL inputs are processed into environmental grades.
- **Start Button:** Prominent call-to-action button that takes the user directly to the primary live Analysis view.

5.4.2 Login Page:

A secure login interface where returning users can re-verify their accounts to continue tracking their long-term environmental metrics.

Controls:

- **Email Textbox:** Clear data input field for the shopper's registered profile email address.
- **Password Textbox:** Masked security text field that safely masks character input during credentials submission.
- **Login Button:** Submits credentials to the FastAPI backend layer to obtain an authenticated session token.

Constraints:

- **Valid Email Required:** The system checks syntax structures via regular expressions before opening network requests.
- **Password Mandatory:** Form fields block submission automatically if the security text block is left blank.

5.4.3 Signup Page:

An account registration interface designed to onboard new users into the platform's sustainability tracking community.

Controls:

- **Name Field:** Input bar collecting the consumer's full name to personalize their impact dashboard.
- **Email Field:** Primary communication input string used as the profile identifier link.
- **Password Field:** Input bar where users establish a unique account password key.
- **Confirm Password:** Verification field checking string alignment to prevent entry errors before database encryption.
- **Register Button:** Submits user parameters to trigger the account creation workflow

5.4.4 Detection Page:

The core interface where users submit e-commerce links to initiate web scraping routines and run AI eco-scoring calculations.

Controls:

- **Large Text Input Area:** Central input bar optimized for pasting e-commerce product links, URLs, or raw material descriptions.
- **Analyze Button:** Submits the target link to trigger backend scraping scripts and the RoBERTa evaluation pipeline.
- **Upload Text Option:** File drop selector that allows users to upload local product text catalogs or material sheets.

Features

- **Real-Time Analysis:** Displays an animated processing wheel to indicate progress while background script modules evaluate material components.
- **Prediction Display:** Temporarily mounts text metrics to show identified fabric compounds and regional manufacturing data points.

5.4.5 Result Page:

A balanced, side-by-side dashboard that displays the final environmental evaluation metrics clearly.

Displayed Data:

1. **Sustainability Label:** A clear color-coded classification badge grading the product's environmental impact (e.g., *Highly Sustainable*, *Greenwashing Risk*).
2. **Eco-Score / Confidence Rating:** A precise radial gauge showing the calculated environmental grade out of 100, paired with an NLP validation score.
3. **Material Analysis:** A broken-down data grid detailing the specific ecological weights of identified fabric materials and packaging materials.
4. **Alternative Recommendations:** A curated list of matching, high-scoring eco-friendly replacement items found in the database catalog.

5.4.6 History Page:

Provides users with a structured timeline of their past activity, encouraging continuous sustainable shopping patterns.

Features:

- **Previous Analyses:** A chronological list detailing past scanned product items, their generated eco-scores, and calculation dates.
- **Search Functionality:** An integrated text filter to quickly locate previously evaluated store links or specific retail items.

- **Filter Options:** Dropdown sorting controls that filter records by product department, store name, or minimum environmental grade.

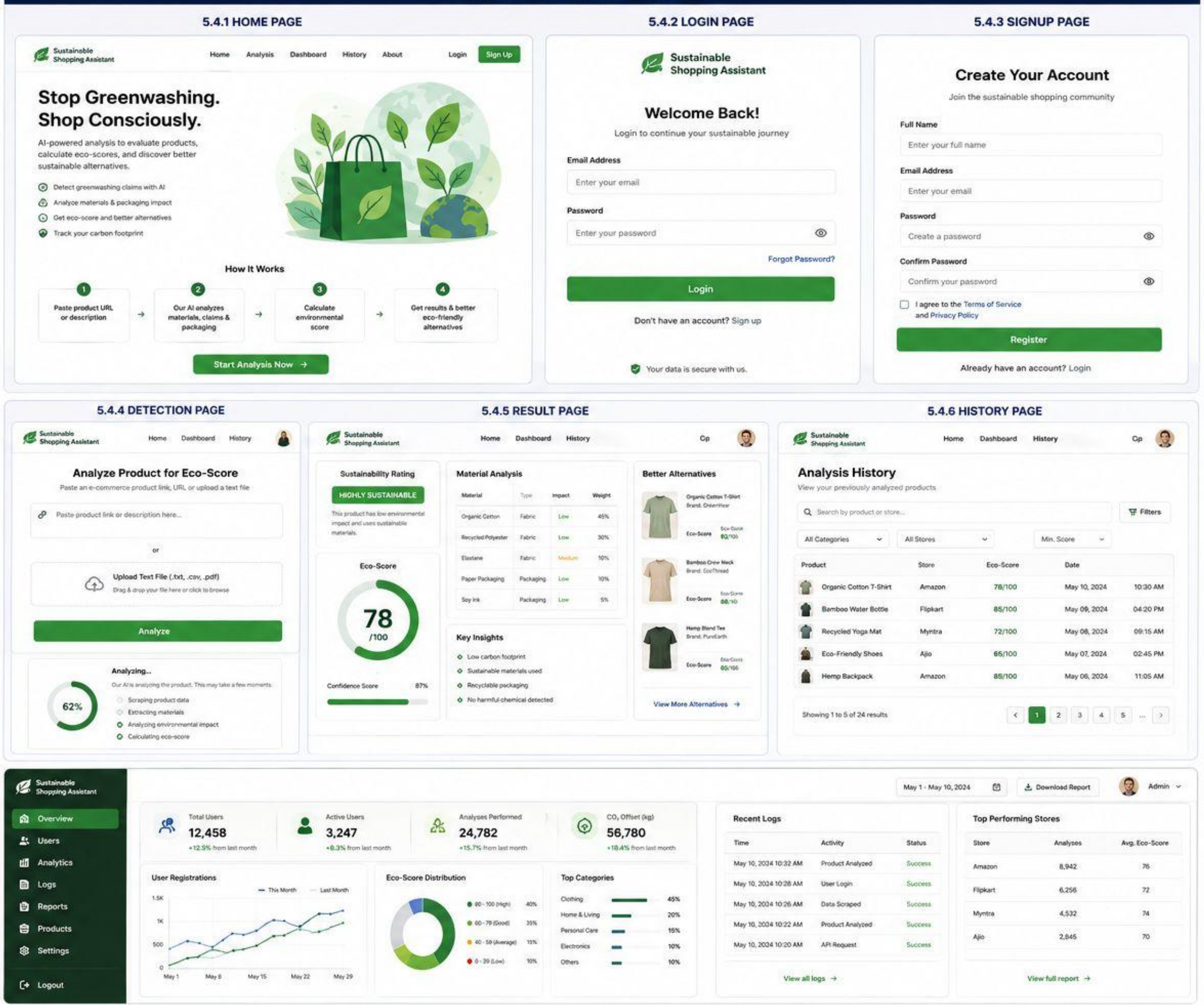
5.4.7 Admin Dashboard:

An analytical oversight panel giving authorized system managers deep performance metrics over the platform.

Features:

1. **User Monitoring:** Tracks active shopper metrics, profile registration rates, and user retention trends over time.
2. **Analytics:** Aggregates macro sustainability metrics, highlighting global carbon footprint offsets achieved by the community.
3. **Logs Management:** Displays live server transactions, detailing API processing loads and active web scraping queues.
4. **Report Management:** Provides tools to oversee the sustainable replacement catalog and verify newly indexed product listings.

5.4 USER INTERFACE SCREENS



Conclusion:

The **Sustainable Shopping Assistant** delivers a reliable software system designed to track product manufacturing contexts and protect consumers against corporate greenwashing. By coupling asynchronous REST API routers (**FastAPI**) with advanced deep learning classification frameworks (**RoBERTa**), the platform converts complex, unstructured e-commerce text descriptions into actionable environmental insights. The completed software design pattern ensures scalable deployment, helping modern consumers confidently transition toward ethical, sustainable, and carbon-conscious purchasing habits.

References:

1. **MongoDB Official Documentation:** Used for designing database architecture, structuring product sustainability history, and managing NoSQL document collections.
2. **RoBERTa: A Robustly Optimized BERT Pretraining Approach (arXiv)** Used for implementing Transformer-based text classification, extracting linguistic material features, and understanding context in product data.
3. **RoBERTa Research Paper (Hugging Face Papers):** Applied for enhancing contextual deep learning patterns and optimizing binary/multi-class classification tasks for brand reliability.
4. **FastAPI Framework Information:** Utilized for structuring the backend application layer, developing rapid e-commerce scraping APIs, and managing async client requests.
5. **SemEval-2024 Research on Machine-Generated and Semantic Text: Analysis** Consulted for fine-tuning text models to identify marketing bias, misleading green claims, and descriptive inconsistencies.
6. **Open-Source Environmental & E-Commerce Datasets:** Used for gathering diverse human-vetted product catalogs, eco-indices, and carbon footprint baseline materials for training matrices.
7. **Hugging Face Transformers Library:** Used for deploying, loading, and optimizing weights of pre-trained language models on domain-specific datasets.
8. **NLTK (Natural Language Toolkit)** :Applied during the data cleaning phase for foundational preprocessing tasks like word tokenization, removing stop-words, and sentence lemmatization.
9. **spaCy NLP Library:**Used for Named Entity Recognition (NER) to isolate material composition percentages and location tags from unstructured text strings.